

Biomaterials: Materials in Medicine (BM 4090/5090)

Assignment 1

15 Questions | Total: 20 Marks

1. For SC, BCC and FCC crystals, identify the crystallographic direction along which nearest-neighbour atoms touch. Using this information, derive the relation between lattice parameter and atomic radius for each structure. A formula without the geometrical justification will not receive marks. (2 mark)
2. The enthalpy of atomization of diamond is 713 kJ mol⁻¹. The listed bond energy is 347 kJ mol⁻¹. The heat of dissociation of the chlorine molecule is 121.3 kJ mol⁻¹. The Cl-Cl bond energy is 242.4 kJ mol⁻¹. Draw the atomic arrangement in the space and use bond counting and the definition of one mole of atoms to explain quantitatively the apparent factor-of-two discrepancy for both diamond (2 mark).
3. H-bonding is directional in nature. Using an appropriate H-bond energy from your textbook (with a drawing of the possible arrangement of all atoms in the ice structure), estimate the fraction of hydrogen bonds disrupted during the melting of ice. Show the bond-counting basis and comment whether melting destroys the H-bond network completely. (1 marks)
4. The potential energy W of a system of two atoms varies as a function of their distance of separation r as follows:

$$W = -A/r^n + B/r^m$$

Show that at equilibrium with a suitable graphical presentation of potential energy well and calculate the following.

- (i) $r = r_0 = (mB/nA)^{1/(m-n)}$
 - (ii) the energy of attraction is m/n times the energy of repulsion, and
 - (ii) the bond energy $W_0 = A/r_0^n (m-n)/m$. (2 marks)
5. The potential energy W for the formation of a bond between two univalent ions is given by the following equation

$$W = \frac{-A' z_1 z_2 e^2}{r} + \frac{B}{r^m} + \Delta E$$

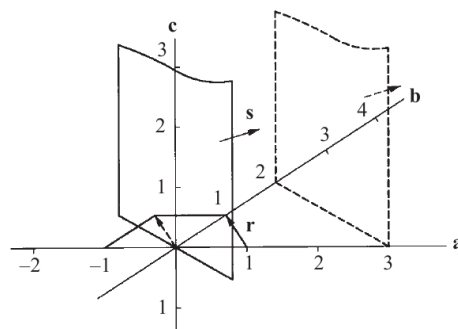
For the given ionic-bond potential with ($r_0=2.50 \text{ \AA}$) and ($m=9$), determine (B) from the equilibrium criterion. Then state how the equilibrium separation would shift if (B)

increased, with all other parameters unchanged. Use a suitable sketch and complete the derivation steps for your answer. (2 marks)

6. Using the tabulated bond energies, estimate the enthalpy of atomization of Cu and W. Derive the number of bonds contributed per atom from their crystal structures, and explain why W has the higher value. (1 marks).

Metal		Bond energy, kJ mol^{-1} (of bonds)
Copper	Cu	56.4
Silver	Ag	47.5
Gold	Au	60.0
Aluminium	Al	54.0
Nickel	Ni	71.6
Platinum	Pt	94.3
Lead	Pb	32.5
Cobalt	Co	70.8
Magnesium	Mg	24.6
Cadmium	Cd	18.6
Zinc	Zn	21.9
Sodium	Na	27.0
Tungsten	W	212.3
Iron	Fe	104.0
Chromium	Cr	99.3
Molybdenum	Mo	164.6
Vanadium	V	128.5
Niobium	Nb	180.8

7. Find the Miller indices of the direction r and the plane indicated by unit normal s in the following figure: (2 marks)



8. Draw two adjacent FCC unit cells and join the top and bottom face centres of each cell. Also join these four face centres to the nearest common corners.
- What is the unit cell outlined by this procedure?
 - Is it in the Bravais list?
 - If so, the FCC cell with a higher symmetry must be a special case of the outlined cell. Explain. (2 marks)

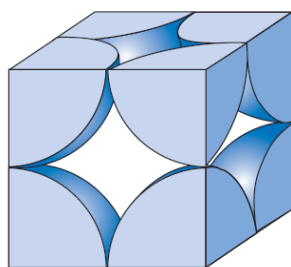
9. Draw a:

- (i) (111) plane in a triclinic lattice.
- (ii) (120) plane in a tetragonal lattice.
- (iii) [11 1] direction in an orthorhombic lattice; and
- (iv) [211] direction in a cubic lattice. (2 marks)

10. Explain why the bond lengths in the following table increase with increasing atomic number, despite the increase in the bond energy. (1 mark)

Crystal	Symbol	No. of bonding neighbours of an atom	Bond energy, kJ mol^{-1}	Bond length, Å	Melting point, $^{\circ}\text{C}$	Boiling point, $^{\circ}\text{C}$
Helium	He	12	0.29	3.57	-270	-269
Neon	Ne	12	0.30	3.15	-249	-246
Argon	Ar	12	1.09	3.76	-189	-186
Krypton	Kr	12	1.50	3.99	-157	-152
Xenon	Xe	12	2.10	4.34	-112	-108

11. Some hypothetical metal has the simple cubic crystal structure shown in Figure. If its atomic weight is 74.5 g/mol and the atomic radius is 0.145 nm, compute its density with step-by-step calculation and derivation. (1 marks)



12. Here are three different crystallographic planes for a unit cell of a hypothetical metal. The circles represent atoms: (a) To what crystal system does the unit cell belong? (b) What would this crystal structure be called? (c) If the density of this metal is 18.91 g/cm³, determine its atomic weight with step-by-step calculation and derivation (2 marks)

